

# **Satellite Structural Inspection Interferometric Synthetic Aperture Radar (InSAR) Based**

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## METHODOLOGY

### 1. Satellite SAR System with Resolution and Revisit Time

To select a suitable SAR system, technical specifications such as spatial resolution, frequency band, polarization, and revisit time were considered. Sentinel-1, operating at C-band (5.405 GHz), was selected due to:

| Parameter          | Sentinel-1 C-SAR (IW Mode)      |
|--------------------|---------------------------------|
| Frequency Band     | C-band (~5.405 GHz)             |
| Revisit Time       | 12 days (single satellite)      |
| Spatial Resolution | 5 m (range) × 20 m (azimuth)    |
| Coverage Mode      | Interferometric Wide Swath (IW) |

### 2. Retrieving Multi-Temporal SAR Imagery Over the Specified Period

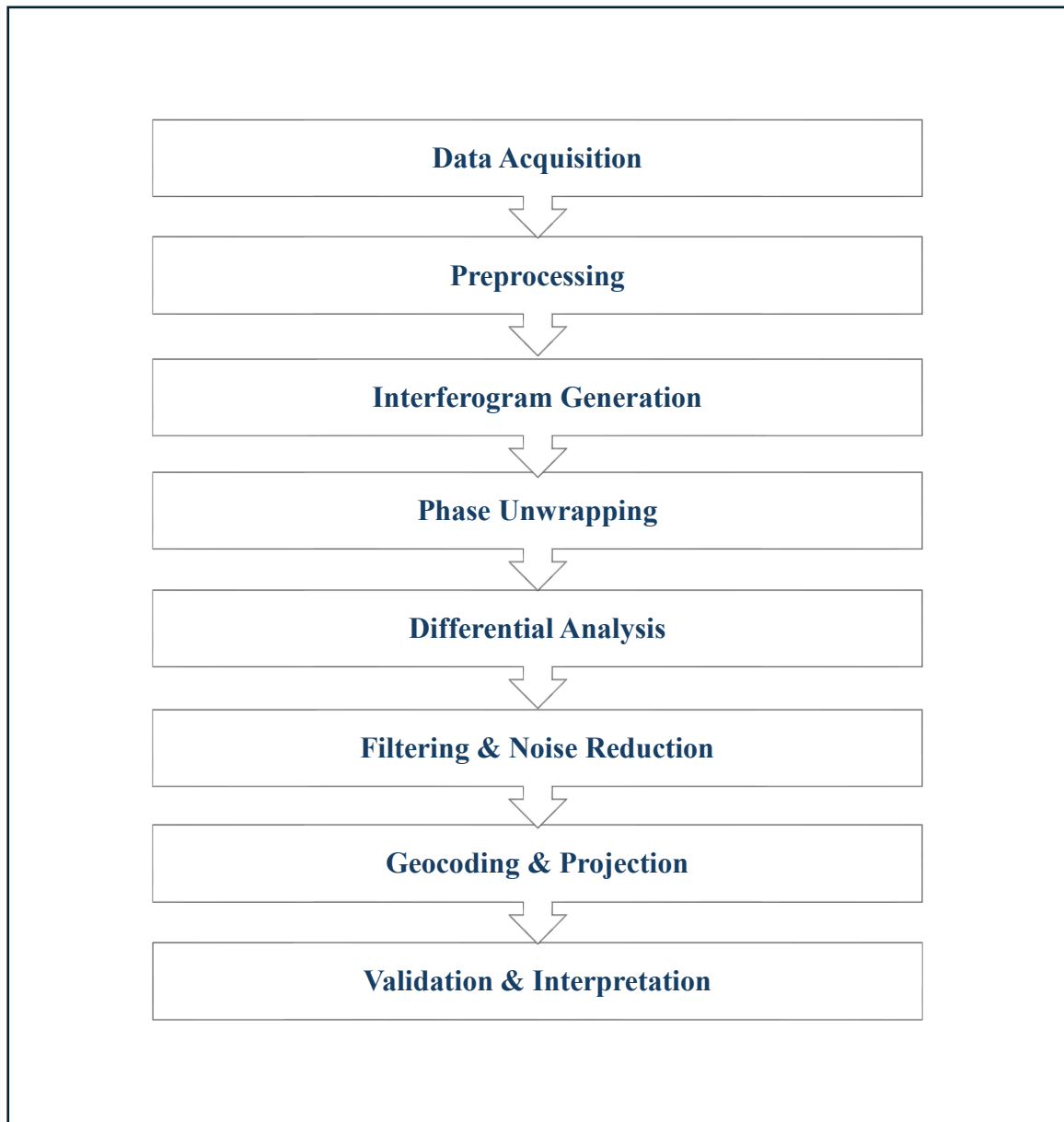
Multi-temporal Sentinel-1 SLC IW mode data were downloaded based on:

- **Area of Interest (AOI):** A polygon covering the bridge and surrounding terrain.
- **Date range:** Multiple acquisitions were collected spanning several months to capture temporal variations.
- **Image acquisition conditions:** All images were selected with the same relative orbit, pass direction (ascending or descending), and polarization (typically VV/VH) to ensure consistency for interferometric analysis.

|                                                                     |
|---------------------------------------------------------------------|
| S1A_IW_SLC__1SDV_20250111T130240_20250111T130310_057397_0710AB_813D |
| S1A_IW_SLC__1SDV_20250123T130239_20250123T130309_057572_07179E_5431 |
| S1A_IW_SLC__1SDV_20250204T130238_20250204T130308_057747_071E86_3FE6 |
| S1A_IW_SLC__1SDV_20250216T130238_20250216T130308_057922_072592_C312 |
| S1A_IW_SLC__1SDV_20250228T130238_20250228T130307_058097_072CB4_7A14 |
| S1A_IW_SLC__1SDV_20250312T130238_20250312T130308_058272_0733CA_217F |
| S1A_IW_SLC__1SDV_20250324T130238_20250324T130308_058447_073AB5_B445 |
| S1A_IW_SLC__1SDV_20250405T130238_20250405T130308_058622_0741CE_2B52 |
| S1A_IW_SLC__1SDV_20250417T130238_20250417T130308_058797_0748F4_3FFD |
| S1A_IW_SLC__1SDV_20250429T130239_20250429T130309_058972_07501E_08D8 |
| S1A_IW_SLC__1SDV_20250511T130238_20250511T130308_059147_0756BA_DE13 |
| S1A_IW_SLC__1SDV_20250523T130238_20250523T130308_059322_075CCB_14B2 |
| S1A_IW_SLC__1SDV_20250604T130237_20250604T130307_059497_0762D9_3AB5 |
| S1A_IW_SLC__1SDV_20250616T130237_20250616T130306_059672_0768D1_F57D |
| S1A_IW_SLC__1SDV_20250628T130236_20250628T130306_059847_076EE3_6BFD |

### 3. Differential SAR Interferometry (D-InSAR) for Displacement Measurement

D-InSAR analysis was performed to measure millimeter-scale deformation over time:



## VISUALIZATIONS

### 1. Defining specific bridge location & parameters for satellite SAR based inspection



#### Project & Bridge Details

State – Rajasthan  
Bridge No. – 903/2  
Bridge Chainage – 1010+560

Project - Chittorgarh to Kota  
Zone – RJ 06  
Structure Type – MNB

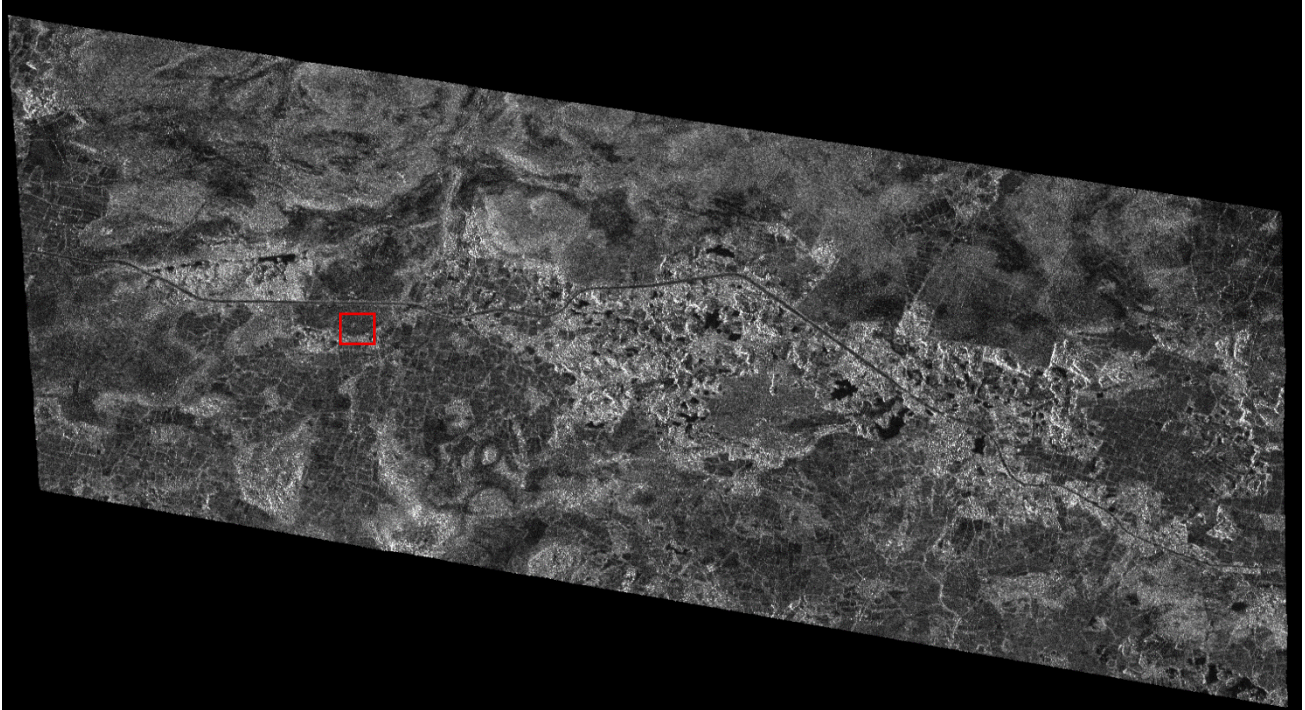
#### Sar Parameters

Sensor – Sentinel-1 (C-Band)  
Beam Mode – IW (Interferometric Wide Swath)  
Path – 34  
Frame – 519

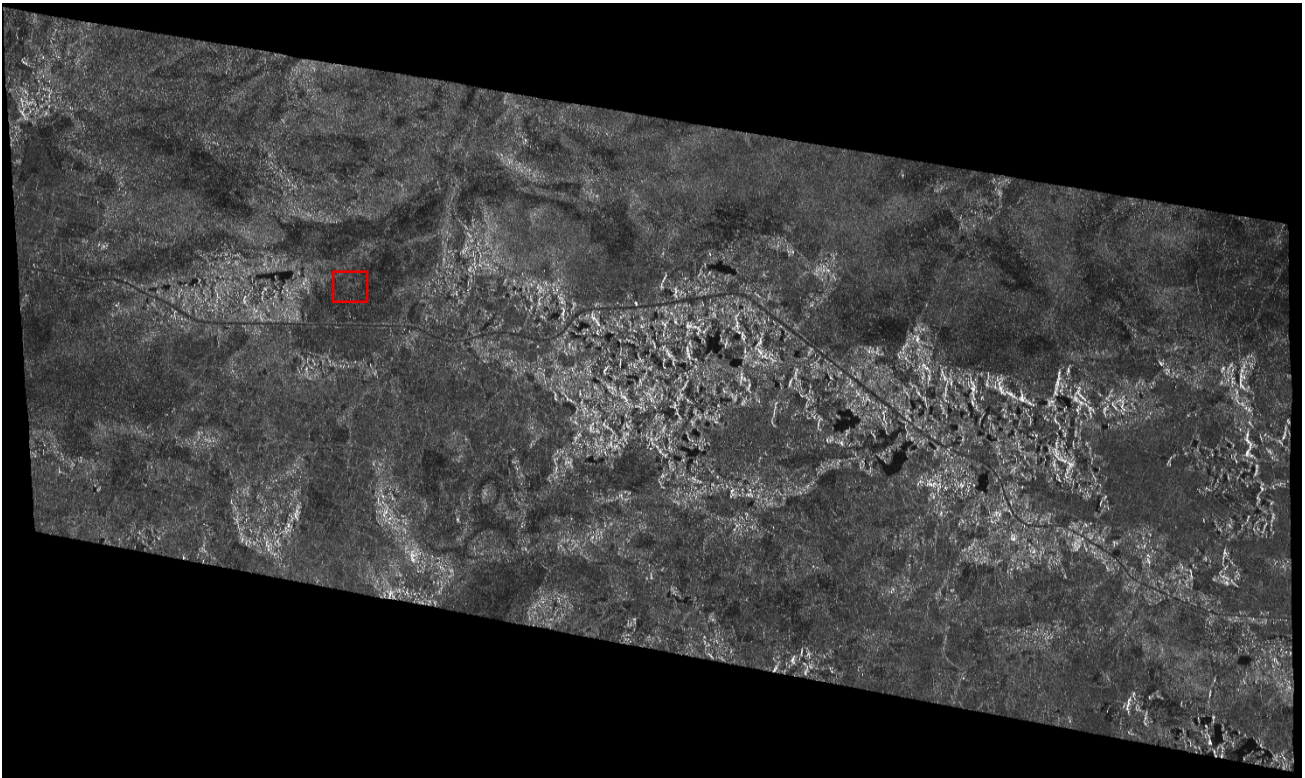
Flight Direction - DESCENDING  
Polarization – VV+VH  
Absolute Orbit – 54706

## 2. Calibration, georeferencing, orthorectification of SAR images (VH/VV)

- Orthorectification of SAR image VH Polarization



- Orthorectification of SAR image VV Polarization

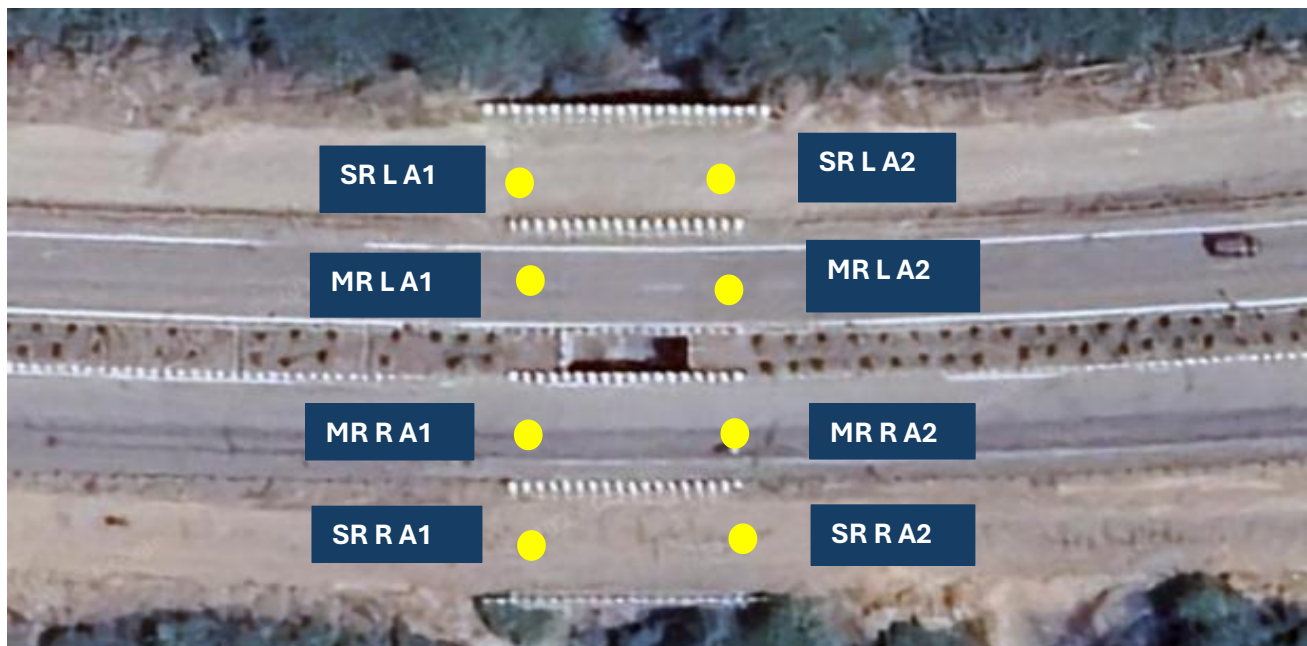


### 3. Results of SAR based bridge inspection, including visualizations

#### Bridge Components

MNB – CH-1010+560

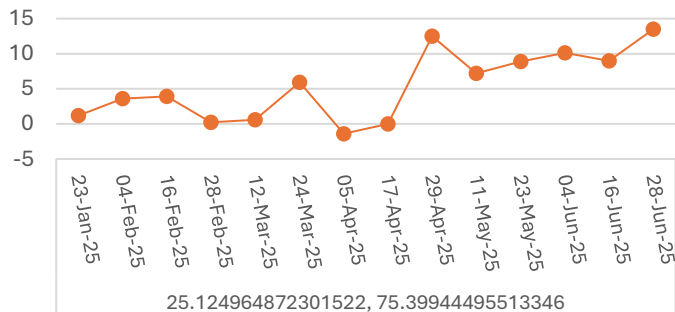
Length – 23.56M



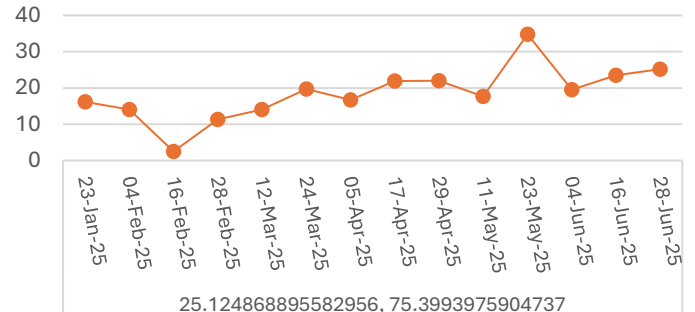
| Scan Duration             |                 | From : 23 Jan 2025 | To : 28 June 2025 |
|---------------------------|-----------------|--------------------|-------------------|
| Components & Abbreviation |                 |                    |                   |
| A1                        | Abutment 1      |                    |                   |
| P1                        | Pier 1          |                    |                   |
| P2                        | Pier 2          |                    |                   |
| A2                        | Abutment 2      |                    |                   |
| LHS                       | Left Hand Side  |                    |                   |
| RHS                       | Right Hand Side |                    |                   |

### 3.1 VV Displacement From 23 Jan 2025 To 28 June 2025

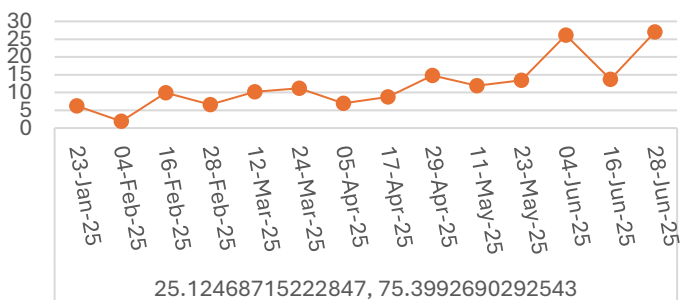
**Displacement @ SR A1 LHS VV**



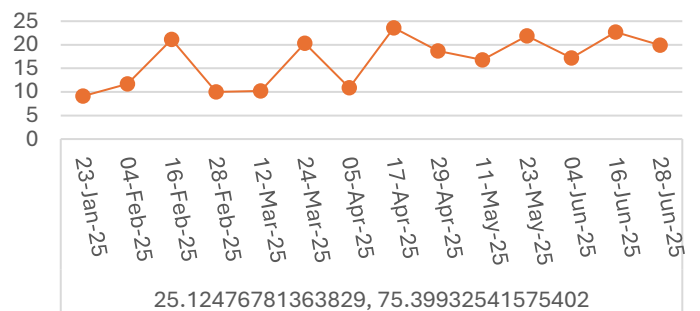
**Displacement @ MR A1 LHS VV**



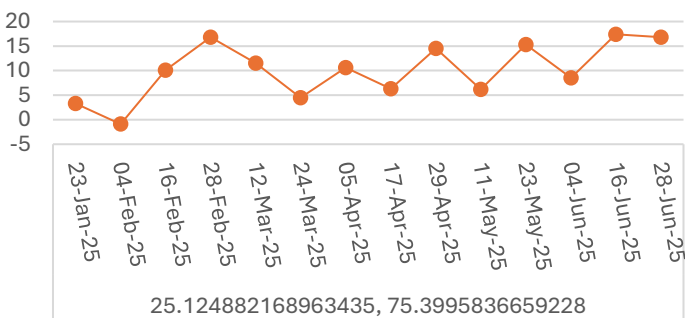
**Displacement @ MR A1 RHS VV**



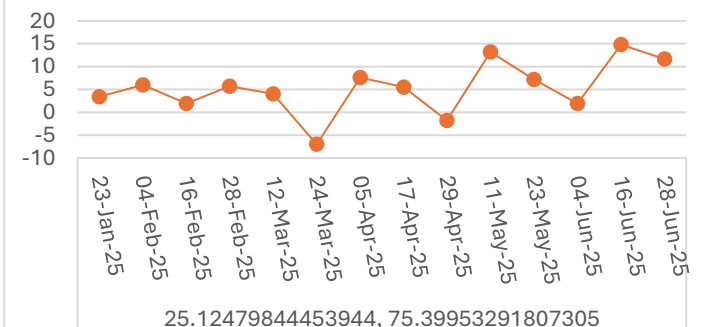
**Displacement @ SR A1 RHS VV**



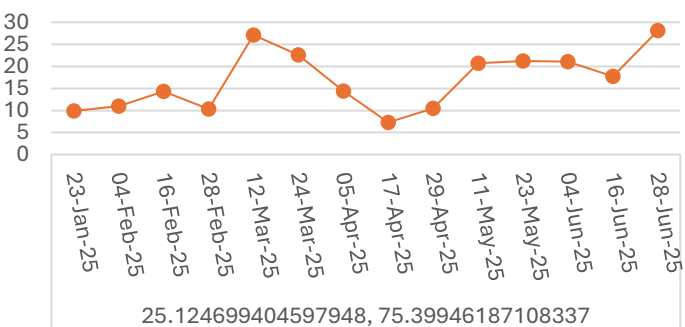
**Displacement @ SR A2 LHS VV**



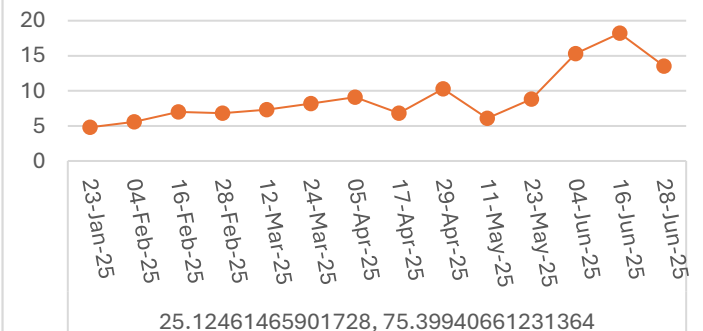
**Displacement @ MR A2 LHS VV**



**Displacement @ MR A2 RHS VV**

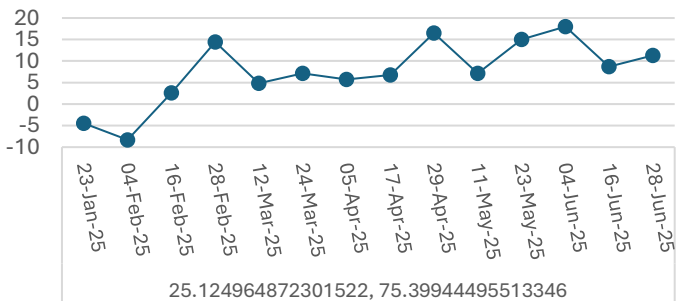


**Displacement @ SR A2 RHS VV**

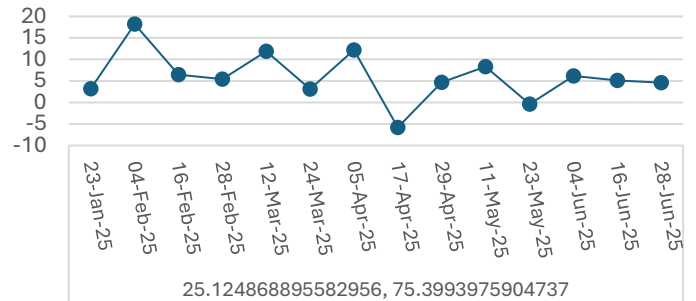


### 3.2 VH Displacement From 23 Jan 2025 To 28 June 2025

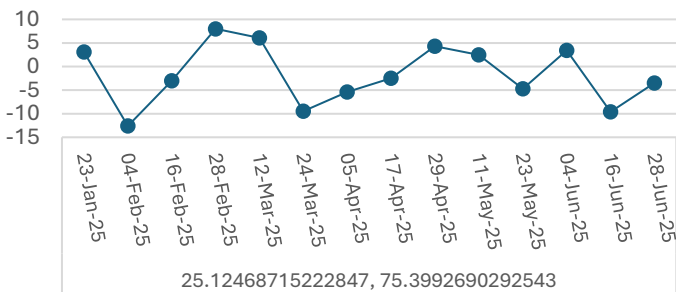
**Displacement @ SR A1 LHS VH**



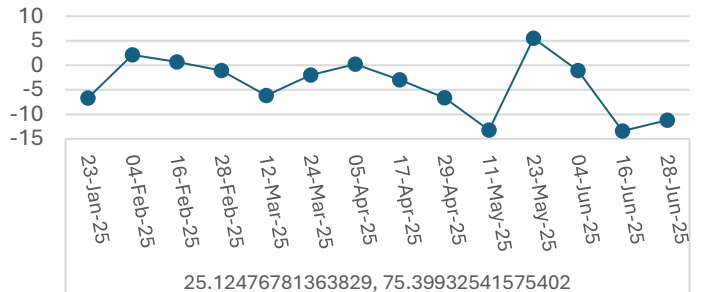
**Displacement @ MR A1 LHS VH**



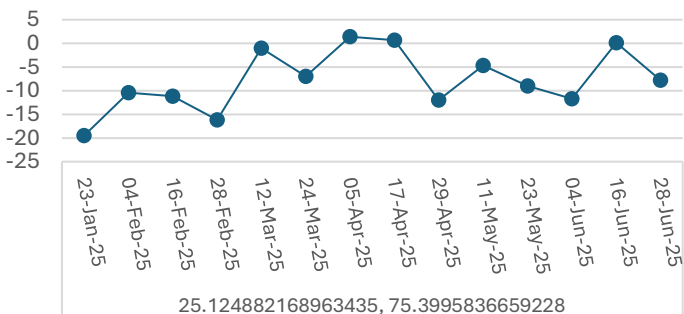
**Displacement @ MR A1 RHS VH**



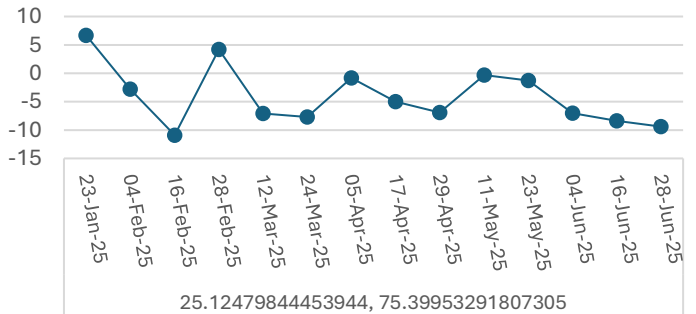
**Displacement @ SR A1 RHS VH**



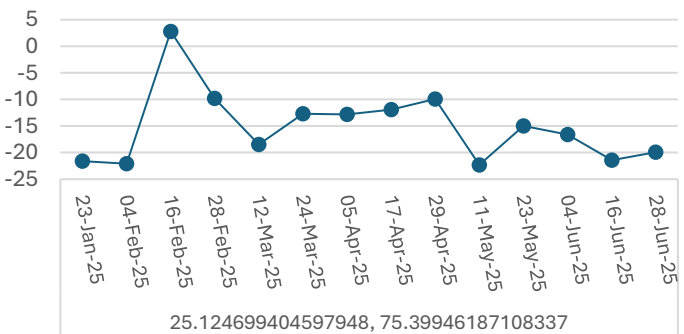
**Displacement @ SR A2 LHS VH**



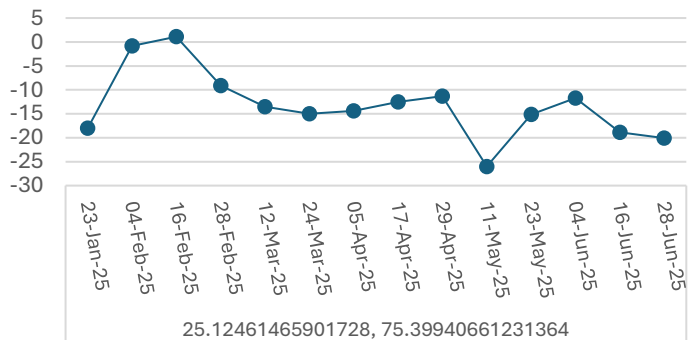
**Displacement @ MR A2 LHS VH**



**Displacement @ MR A2 RHS VH**



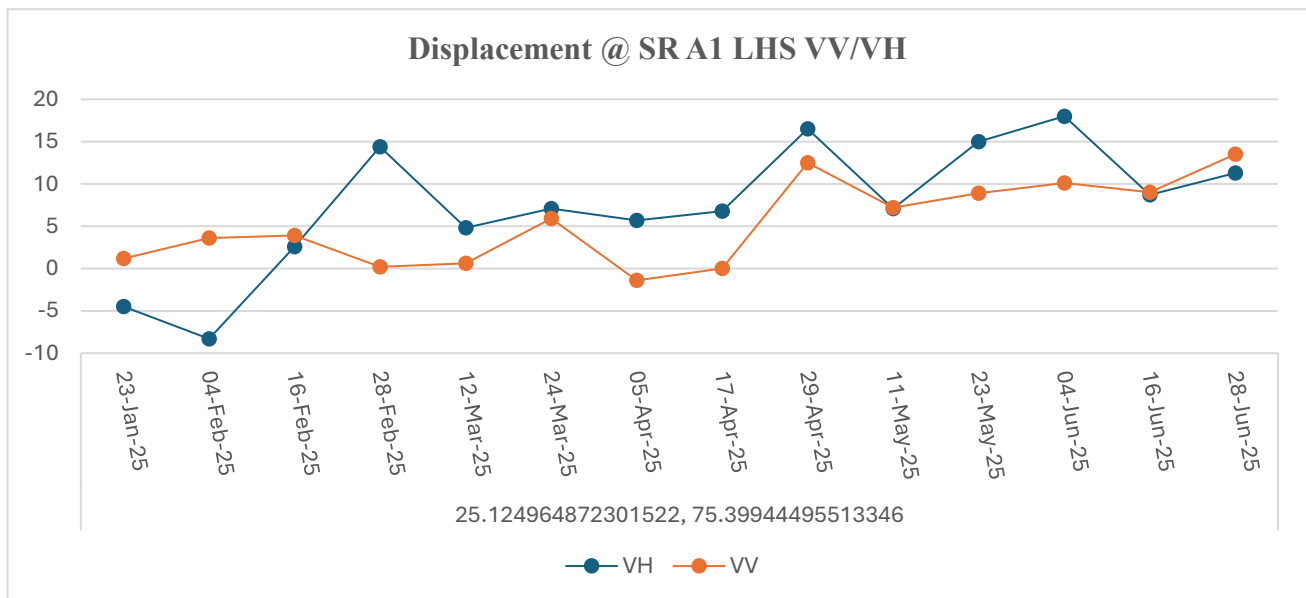
**Displacement @ SR A2 RHS VH**



### 3.3 SAR Based Analysis at Service Road Abutment 1 LHS:



**SR LHS A1** (25.124964872301522, 75.39944495513346)



#### Key Observations:

##### Displacement Magnitude Patterns:

- **Vertical Displacement (VV)** Vertical displacement remains predominantly positive with moderate variability over the monitoring period (Jan–Jun 2025).
- **Horizontal displacement (VH)** Horizontal displacement exhibits greater variability, ranging approximately between –8 mm and +18 mm.

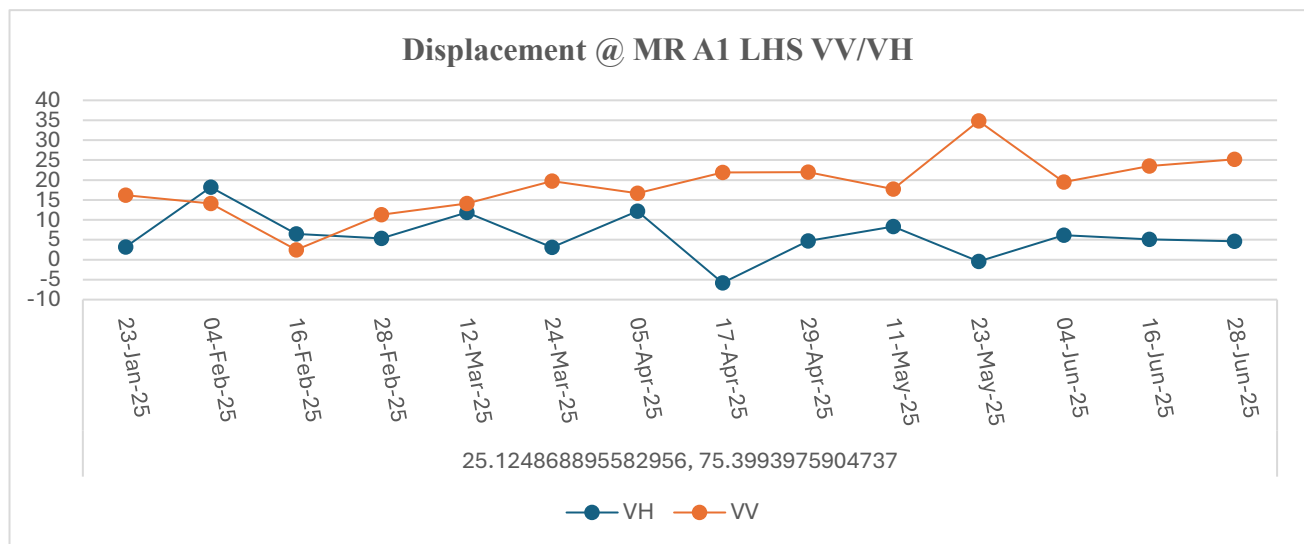
##### Critical Events & Anomalies:

- **February 2025:** VH records a notable negative dip (–8 mm) while VV remains slightly positive, indicating a short-term lateral reversal without vertical settlement.
- **April 2025:** Both VH and VV increase simultaneously (VH +16–17 mm, VV +12–13 mm), suggesting a phase of combined lateral and vertical displacement intensification.
- **June 2025:** VH reaches its maximum observed value (+18 mm) while VV remains elevated (+10 mm), indicating continued lateral dominance.

### 3.4 SAR Based Analysis at Main Road Abutment 1 LHS:



MR LHS A1 (25.124868895582956, 75.3993975904737)



#### Key Observations:

##### Displacement Magnitude Patterns:

- **Horizontal displacement (VH)** Horizontal displacement shows moderate variability, oscillating between approximately -6 mm and +12 mm.
- **Vertical displacement (VV)** Vertical displacement exhibits high variability with a strong upward bias throughout the monitoring period.

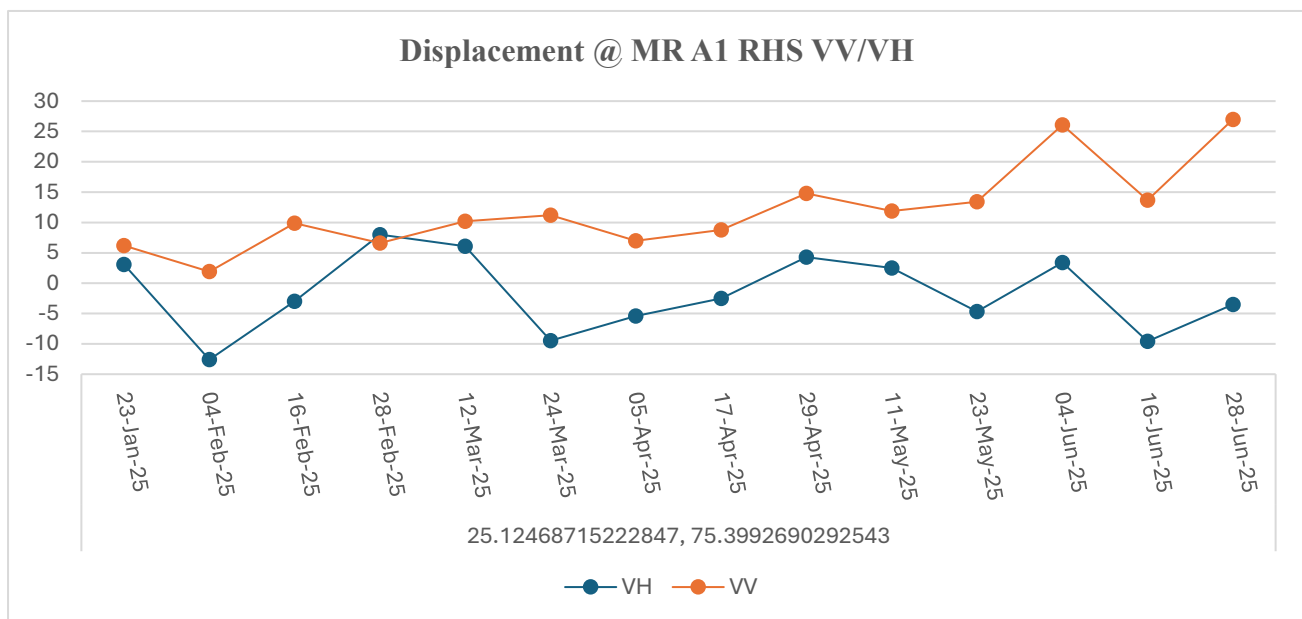
##### Critical Events & Anomalies:

- **February 2025:** VH reaches a local positive peak (+12 mm), while VV remains moderately elevated, indicating an early lateral response phase.
- **April 2025:** VH drops to its maximum negative value (-6 mm), contrasting with rising VV values (+20–22 mm), suggesting decoupled lateral and vertical responses.
- **May 2025:** VV records its highest uplift (+34–35 mm), representing the most significant vertical movement observed across the monitoring period, while VH remains relatively restrained.

### 3.5 SAR Based Analysis at Main Road Abutment 1 RHS:



MR RHS A1 (25.12468715222847, 75.3992690292543)



#### Key Observations:

##### Displacement Magnitude Patterns:

- **Vertical Displacement (VV)** Vertical displacement demonstrates moderate to high uplift with a clear increasing trend over the monitoring period.
- **Horizontal Displacement (VH)** Horizontal displacement exhibits predominantly negative behaviour, indicating sustained lateral movement in a dominant direction.

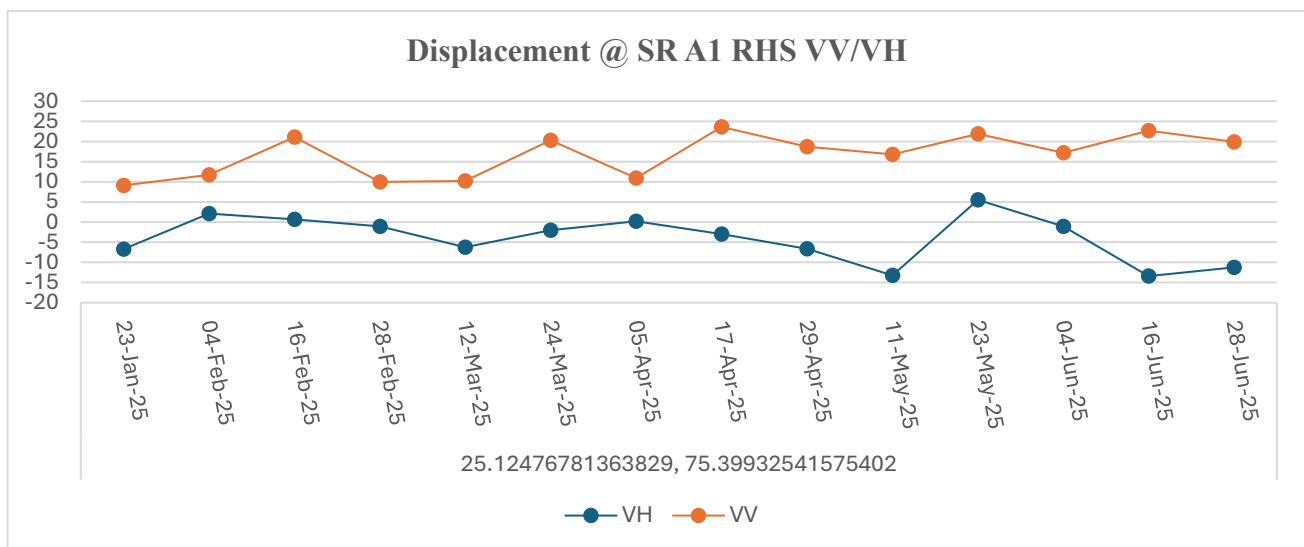
##### Critical Events & Anomalies:

- **February 2025:** VH reaches a sharp negative dip (–13 mm) while VV remains low-positive, suggesting a short-term lateral disturbance.
- **April 2025:** VH recovers briefly to around 0 to +2 mm, coinciding with increasing VV (+14–15 mm), indicating temporary lateral stabilisation.
- **June 2025:** VV records its highest uplift values (+25–28 mm) while VH again trends negative (–8 to –12 mm), highlighting dominant vertical movement with continued lateral bias.

### 3.6 SAR Based Analysis at Service Road Abutment 1 RHS:



SR RHS A1 (25.12476781363829, 75.39932541575402)



#### Key Observations:

##### Displacement Magnitude Patterns:

- **Vertical Displacement (VV)** Vertical displacement remains consistently positive and moderately high throughout the monitoring period (Jan–Jun 2025).
- **Horizontal Displacement (VH)** Horizontal displacement is predominantly negative, indicating sustained lateral movement in a dominant direction.

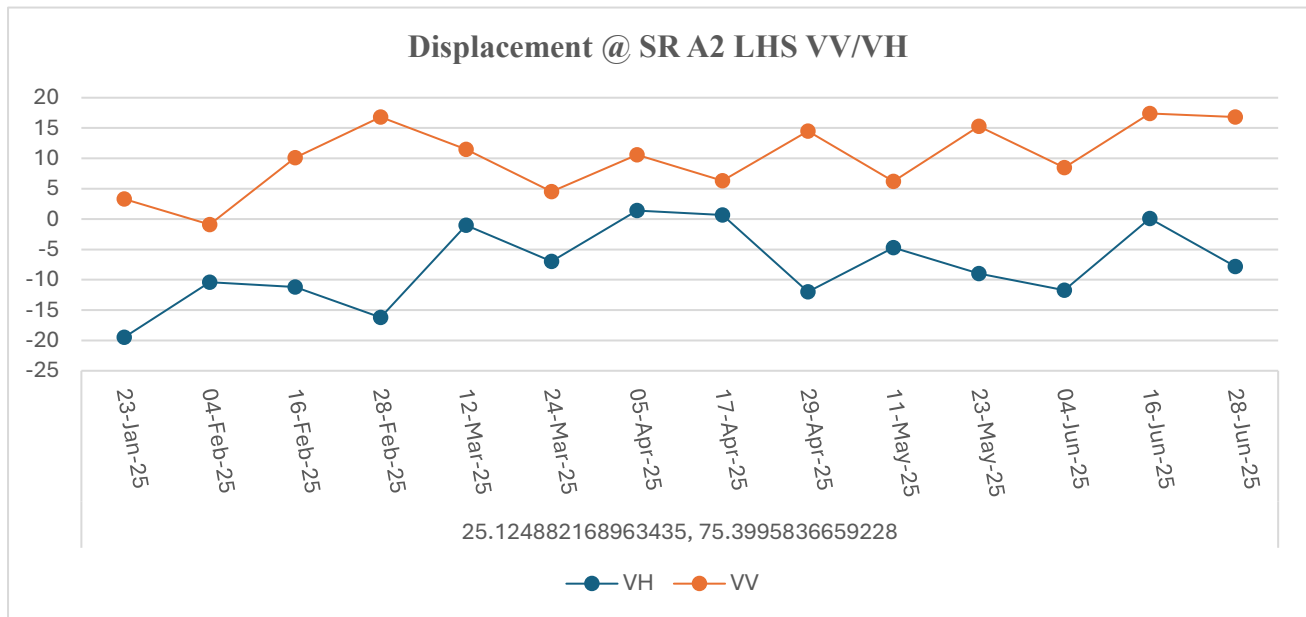
##### Critical Events & Anomalies:

- **February 2025:** VV reaches a local uplift peak (+20 mm) while VH remains moderately negative, indicating vertical dominance without lateral intensification.
- **May 2025:** VH briefly recovers toward 0 mm, while VV remains elevated (+20 mm), indicating a short phase of lateral stabilisation.
- **June 2025:** VH drops again to –15 to –18 mm, while VV increases, marking renewed lateral movement under sustained vertical uplift.

### 3.7 SAR Based Analysis at Service Road Abutment 2 LHS:



SR LHS A2(25.124882168963435, 75.3995836659228)



#### Key Observations:

##### Displacement Magnitude Patterns:

- **Horizontal Displacement (VH)** Horizontal displacement is predominantly negative, indicating sustained lateral movement in a dominant direction.
- **Vertical Displacement (VV)** Vertical displacement remains consistently positive with moderate variability throughout the monitoring period (Jan–Jun 2025). VV values generally range between +2 mm and +18 mm.

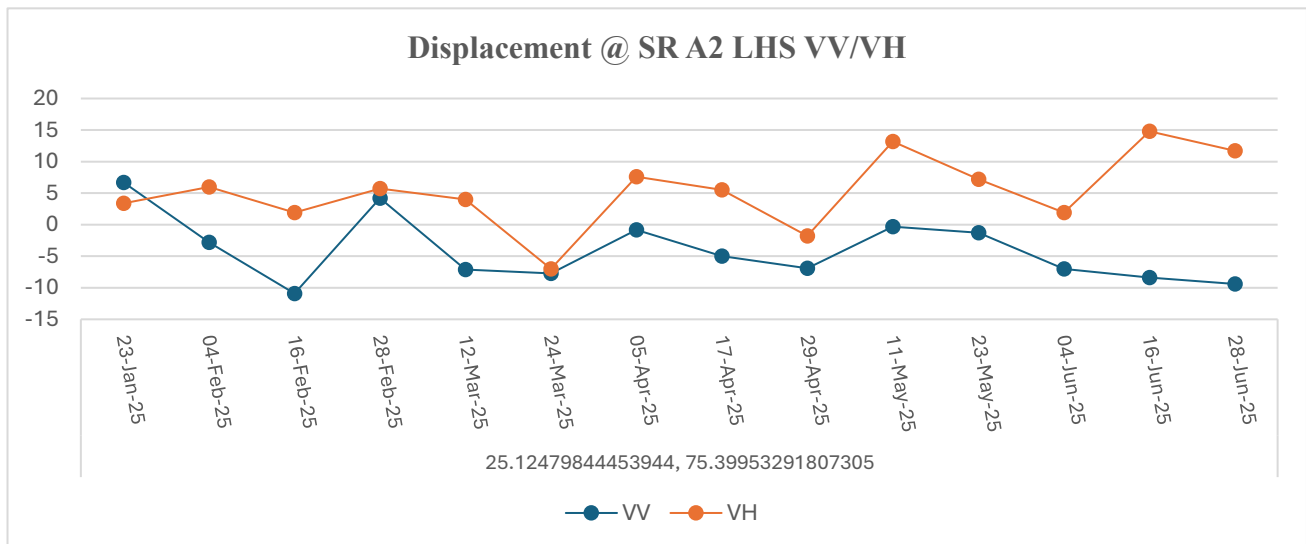
##### Critical Events & Anomalies:

- **February 2025:** VH reaches a significant negative peak (–20 to –22 mm) while VV records elevated uplift (+15–17 mm), suggesting simultaneous vertical uplift and lateral movement intensification.
- **April 2025:** VH shows temporary recovery toward –5 mm, while VV reduces slightly, indicating a short-lived lateral relaxation phase.
- **June 2025:** VV attains its maximum uplift (+17–18 mm), whereas VH continues to fluctuate at moderate negative levels, reflecting sustained vertical dominance with controlled lateral response.

### 3.8 SAR Based Analysis Main Road Abutment 2 LHS:



MR LHS A2(25.12479844453944, 75.39953291807305)



#### Key Observations:

##### Displacement Magnitude Patterns:

- **Vertical Displacement (VV)** Vertical displacement is predominantly negative, indicating sustained downward movement throughout the monitoring period (Jan–Jun 2025). VV values generally range between –2 mm and –12 mm.
- **Horizontal Displacement (VH)** Horizontal displacement remains mostly positive with moderate variability, fluctuating approximately between +2 mm and +15 mm.

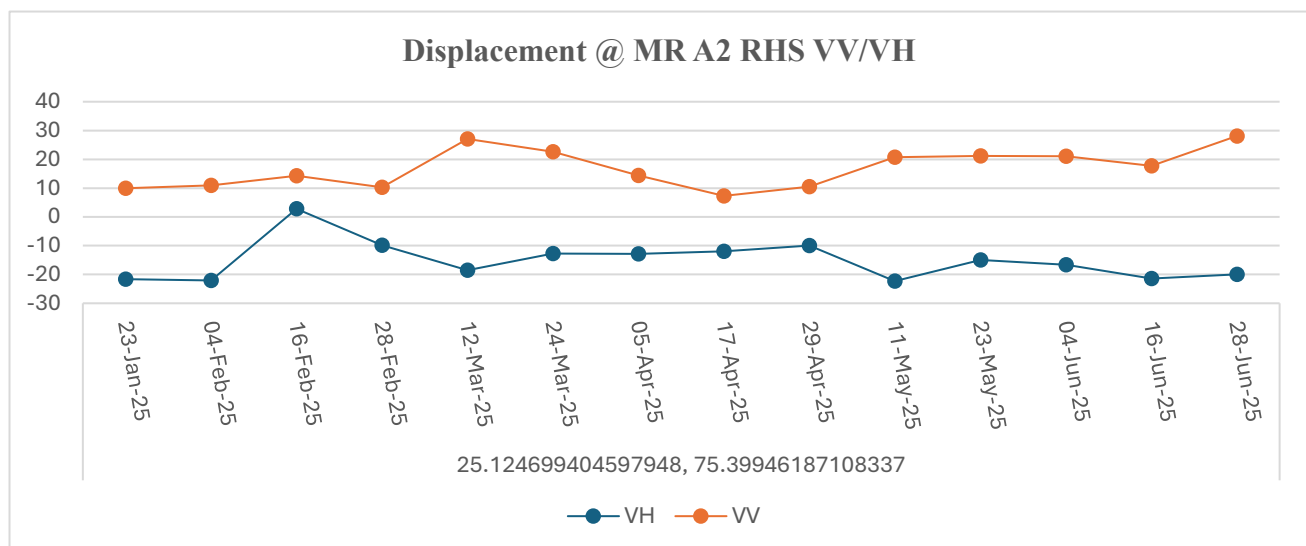
##### Critical Events & Anomalies:

- **February 2025:** VV records a sharp negative dip (–10 to –12 mm), while VH remains positive, indicating a pronounced vertical settlement event without corresponding lateral reduction.
- **June 2025:** VH reaches its highest positive values (+13 to +15 mm), while VV continues to decline, highlighting divergent horizontal uplift and vertical settlement behaviour.

### 3.9 SAR Based Analysis at Main Road Abutment 2 RHS:



**MR RHS A2** (25.124699404597948, 75.39946187108337)



#### Key Observations:

##### Displacement Magnitude Patterns:

- **Vertical Displacement (VV)** Vertical displacement exhibits consistently high uplift with moderate variability throughout the monitoring period (Jan–Jun 2025). VV values generally range between +8 mm and +28 mm.
- **Horizontal Displacement (VH)** Horizontal displacement is predominantly negative, indicating sustained lateral movement in a dominant direction.

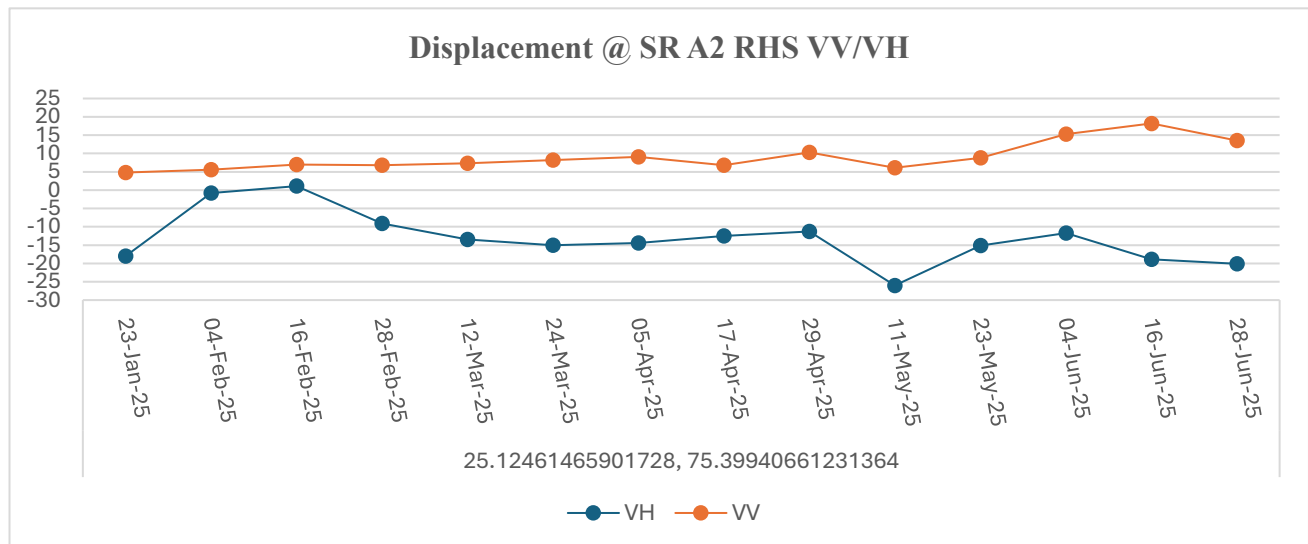
##### Critical Events & Anomalies:

- **February 2025:** VH shows a short-term recovery (–8 to –10 mm), while VV continues rising, indicating temporary lateral relaxation.
- **June 2025:** VV remains elevated and increases further (+20–28 mm), while VH stays strongly negative (–18 to –26 mm), highlighting persistent vertical dominance with sustained lateral bias.

### 3.10 SAR Based Analysis at Service Road Abutment 2 RHS:



SR RHS A2 (25.12461465901728, 75.39940661231364)



#### Key Observations:

##### Displacement Magnitude Patterns:

- **Vertical Displacement (VV)** Vertical displacement remains consistently positive with a gradual upward trend throughout the monitoring period (Jan–Jun 2025). VV values generally range between +4 mm and +20 mm.
- **Horizontal Displacement (VH)** Horizontal displacement is predominantly negative, indicating sustained lateral movement in a dominant direction. VH values fluctuate approximately between –10 mm and –28 mm, with notable variability.

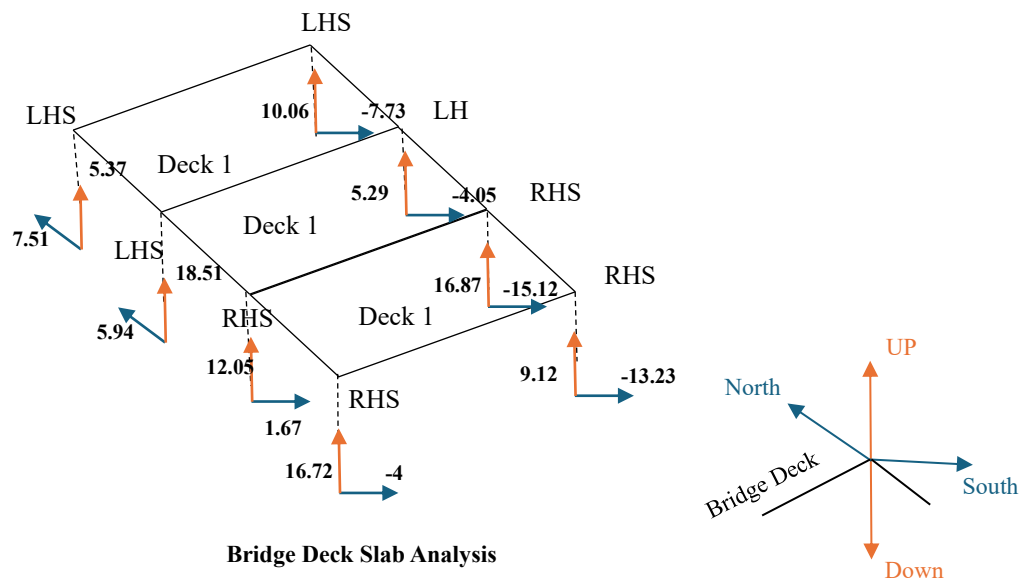
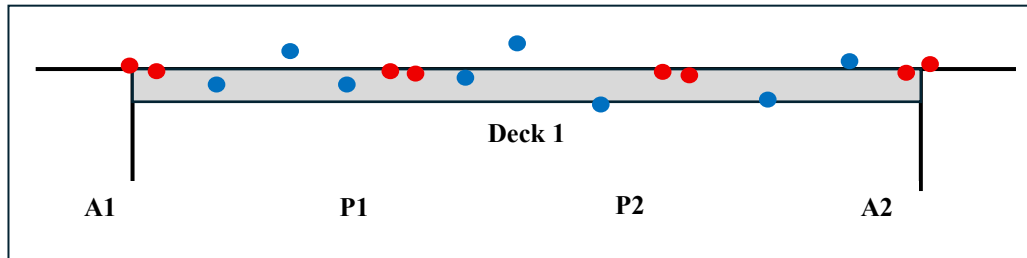
##### Critical Events & Anomalies:

- **Feb 2025:** VH temporarily recovers toward –5 to –7 mm, while VV remains low-positive, indicating short-term lateral relaxation.
- **May 2025:** VH reaches its most negative value (–28 mm), coinciding with VV around +6 mm, representing the most significant lateral displacement event in the dataset.
- **June 2025:** VV attains its highest uplift values (+18–20 mm), while VH remains moderately negative, reflecting dominant vertical uplift with controlled lateral response.

## ANALYSIS

### 1. Identify potential deformation, shifts or movement of the bridge structure

- Points of Analysis (POA) of bridge (●)



#### Vertical Displacement Behaviour:

- The bridge deck slab exhibits non-uniform vertical displacement across both the LHS and RHS regions. The maximum uplift is observed near the lower central deck location, reaching approximately +16.7 mm, while other notable uplift values occur at the central RHS upper deck point (+16.9 mm) and the **upper** LHS deck region (+10.1 mm).

#### Horizontal Displacement Behaviour:

- Horizontal displacement magnitudes show strong directional variability, ranging from approximately +16.7 mm towards the RHS at the lower deck region to about -15.1 mm near the upper RHS deck point. The LHS deck edges experience comparatively smaller but noticeable lateral movement (+5.9 mm to +7.5 mm), while the RHS deck zones exhibit pronounced opposing horizontal displacements (-13.2 mm to -15.1 mm), highlighting significant lateral distortion

#### CONCLUSION:

- The combined vertical and horizontal displacement patterns indicate a highly non-symmetric deformation response of the bridge deck slab. The LHS region is predominantly characterised by moderate uplift with relatively restrained lateral movement, whereas the RHS region demonstrates larger horizontal displacements accompanied by alternating uplift and localized settlement